

PROCEED v6.2

Phenotypic Rates of Change Evolutionary and Ecological Database

Overview

General

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Data Contributions

Carnegie Museum of Natural History, Maine Department of Marine Resources, NS-ITBS surveys (www.ices.dk/marine-data/dataportals/Pages/DATRAS.aspx), Powdermill Avian Research Station

The Phenotypic Rates of Change Evolutionary and Ecological Database (PROCEED) is an ongoing compilation of rates of phenotypic change. PROCEED provides information to compute two popular and complementary evolutionary rates, Darwins and Haldanes, in contemporary timescales. PROCEED also provides a set of biological and methodological variables aimed at modeling and understanding the variability in those rates of phenotypic change.

The Darwins (D) rate of change is the proportional change in a quantitative trait (x) over time (Haldane 1949).

$$D = \frac{1}{x} \cdot \frac{dx}{dt}$$

The integral of this equation across a discrete period ($t=t_2-t_1$) is the difference between the natural logarithm of a trait value at a particular time (or in a particular population) and the natural logarithm of the same trait at some time in the past (or in another population), over the length of time (typically expressed as millions of years).

$$D = \frac{\left[\ln(\bar{x}_2) - \ln(\bar{x}_1) \right]}{\left[t_2 - t_1 \right]}$$

$$D = \frac{\ln(\bar{x}_2/\bar{x}_1)}{t}$$

This means that the value of the trait at time 2 is a function of the value at time 1 as follows:

$$\bar{x}_2 = \bar{x}_1 \cdot e^{D \cdot t}$$

The Haldanes (H) rate of change (Haldane 1949) scales the magnitude of change by the amount of variation in the trait. Lerman (1965) further articulated this measure by expressing rates as "differences between the population means in units of standard deviation." Gingerich (1993) formulated a similar metric and dubbed it the Haldane (the same measure is designated D' H by Lynch 1990). A simple formulation of the Haldane is:

$$H = \frac{\frac{(\bar{x}_2 - \bar{x}_1)}{SD_x}}{g}$$

where $\bar{x}_2$ and $\bar{x}_1$, represent mean trait values for each of two populations (synchronic) or for a single population at two different times (allochronic), SD_x is the pooled standard deviation, and g is the number of generations separating the populations or samples (years divided by generation time). For many traits, especially morphological traits, standard deviations are expected to increase with the mean (i.e., coefficient of variation remains relatively constant). For such traits, raw data should be transformed to natural logarithms (ln), which will reduce heteroscedasticity (Wright 1968). However, estimated rates of change calculated using ln

means and standard deviations do not differ appreciably from those calculated using non-transformed means and standard deviations (Hendry & Kinnison 1999).

Darwins and Haldanes differ in two fundamental ways. First, Darwins specify the rate of proportional change in units of the initial value assuming an exponential rate of change, whereas Haldanes specify the rate of change in standard deviation units. Second, the time interval is measured per millions of years for Darwins and in generations for Haldanes. In principle (but not in practice), these two differences are independent of each other. Indeed, a proportional change could just as easily be specified per generation or a standardized change per year. If an investigator's goal is to measure a change in some organism that is relevant to time dependent human interests, then proportionate change and a time unit based on years could be the most desirable characteristic of a rate measure (i.e., Darwins). However, if the goal is to understand how a population responds to environmental change or to estimate the intensity of selection, then standardizing by the trait's variation and using a time scale more relevant to the organism (i.e., generations) will provide a more appropriate rate measure (i.e., Haldanes).

In accounting for variation, Haldanes have a better grounding in the evolutionary process, but this property adds additional complexity when comparing genetic and phenotypic studies. Environmentally-induced variation might be greater in the wild than it is under common conditions. In such cases, genetic Haldanes will be higher than phenotypic Haldanes for the same absolute magnitude of change. In contrast, if individuals with extreme phenotypes are selected against in the wild, but not in the laboratory, genetic Haldanes may be lower than phenotypic Haldanes for the same magnitude of change.

The accuracy of an estimated evolutionary rate depends on the accuracy of estimated elapsed time (years for Darwins, generations for Haldanes). Estimating the number of years that have passed might seem easy, but it is not without ambiguity in many instances. Small errors in the estimated year of population founding, for example, can translate into measurable differences in evolutionary rates when the total time interval is relatively short. Obtaining good estimates of the time interval in generations is often more difficult because of the added uncertainty in the estimation of generation time. As a result, rates estimated in Haldanes may be less accurate than those estimated in Darwins, regardless of how well each measure reflects the evolutionary process.

The dimension in which a trait is measured (e.g., length, area, volume) influences rates of evolution calculated using Darwins (Gingerich 1993). This dimension dependence arises because as length increases, area increases as the square function of that change, and volume as the cube function of that change (assuming a cubic trait). On the other hand, Haldanes are not dimension dependent. Phenotypic data is commonly measured on ratio or interval scales. Data on a ratio scale have a constant interval between adjacent units, and the measurement scale has a precise zero point corresponding to a null quantity. Ratio scale data have the property that doubling a value doubles the actual quantity. In contrast, data on an interval scale have a constant interval between adjacent units, but the zero point is arbitrary

(e.g., time of day). Darwins are not appropriate for specifying rates of evolution for interval scale traits. Behavioral and phenological traits are often measured on an interval scale, such as migratory timing, or flowering time. Darwins cannot be used to estimate rates in those cases because, for example, in the migration case, What is day zero? If it is assumed to be the first day of the year, the rate of change will be different from the rate of change for the same amount of change if day zero is assumed to be June 1. On the contrary, Haldanes is still suitable for calculating evolutionary rates on interval scales.

In a similar way to the issues with the interval scales, for morphological traits, the choice of the landmarks may impact calculated Darwins if an evolutionary change is not proportional along its entire structure being measured (i.e., the change occurs in a smaller section of the structure). In those cases, the landmark operates as a misleading zero. This discrepancy arises because the same change in only part of the structure will appear smaller when measured from a more distant zero point. Haldanes are not as strongly influenced as Darwins by the choice of different landmark points when part of the trait is invariant. Furthermore, both Darwins and Haldanes are ill-equipped to handle rates of change in nonmetric characters, such as color morphs, presence or absence of a character state, behavioral options, or alternative life histories.

Darwins and Haldanes provide different but complementary measures of evolutionary rate. Because both specify a phenotypic difference per unit time, rates calculated using each will be correlated in general. However, some studies might have relatively low Haldanes rates but high Darwins rates, or the opposite, because of differences in the time units (years vs generations), or because of the unit of change (trait initial value vs trait variance). Due to the different conceptual basis of the two measures and to the different information conveyed by each, it is worth presenting both when possible. It is also important to acknowledge the limitations of each measure as discussed above.

PROCEED includes studies that measure the intraspecific change in quantitative (continuous or discrete) traits and report the time elapsed from the onset of environmental novelty or a reference to a historical or biological event reported in other sources (e.g., a mine opening, a well-documented biological invasion). The maximum elapsed time between the environmental change and the sampling was no longer than 500 years (figure 1). However, most of the observations correspond to less than 100 years of elapsed time (3rd quartile 89.5 years) and only a few cases correspond to more than 200 years of elapsed time (figure 1). The time measured as years and generations were positively correlated, but for a given time in years, there is a wide range of variability in generations, approximately two orders of magnitude (figure 1). The included studies followed a single population through time (allochronic design) or compare two or (more) populations that diverged at a known time (synchronic design). The studies may record the differences among populations in the field (here called phenotypic studies) or in common growing conditions (here called genetic studies). The database records the change in diverse trait types and environmental changes (Table 1, see “Variable information” section for definitions).

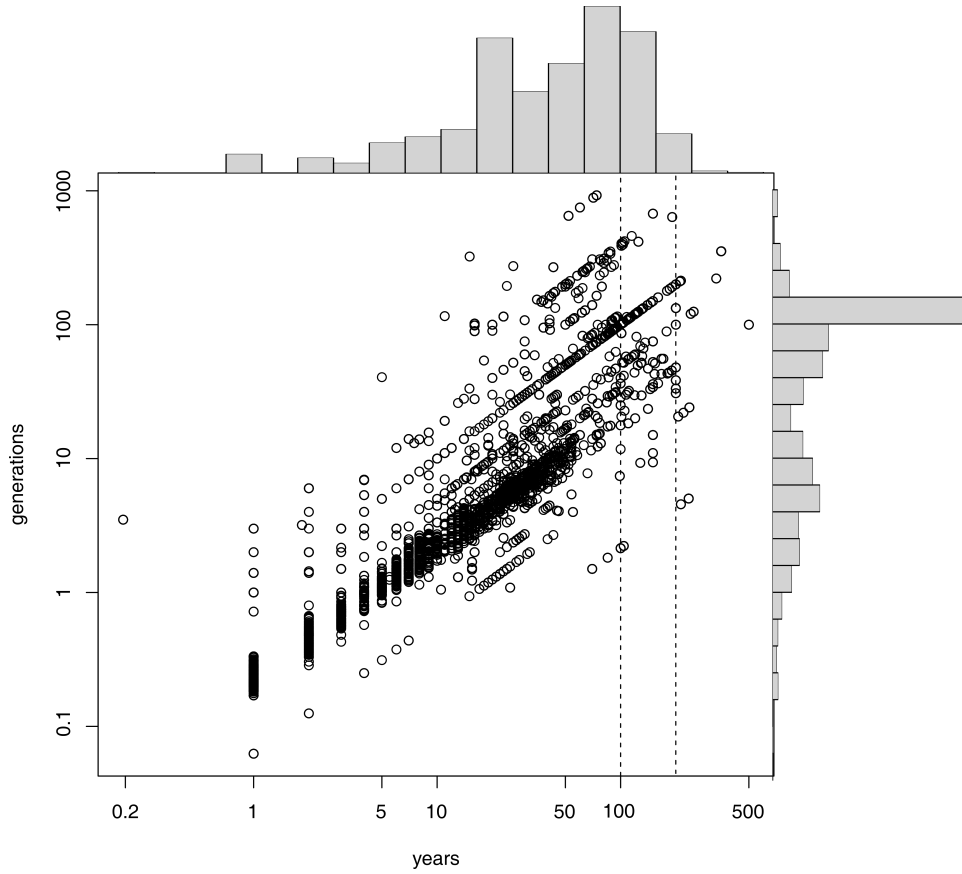


Figure 1. Distribution of elapsed times in PROCEED both, in years and in generations on a log scale. The vertical dashed lines point out 100 and 200 years respectively.

In 1999, a database of phenotypic rates of change in wild populations was compiled. Since then, researchers have used (and expanded) this database to examine the phenotypic responses as a function of the features of the study system (i.e., the population or set of populations, of a given species, that experienced a specific driver or disturbance), the measured traits, and methodological approaches. We compile and add data regularly to the dataset. This dataset is continually being updated as more people include new variables and observations. PROCEED v6.2 is now being released. This last version has 9263 observations, from 1801 worldwide distributed systems (i.e., a population or a set of populations, of a given species, experiencing a specific driver/disturbance) (figure 2), belonging to 428 species. This information was extracted from 326 primary sources of information (i.e. papers, technical reports, etc.). Table 1 shows the distribution of the data in a selection of the classification factors in PROCEEDv6.2_RatesDB. For a full description of the variables and levels see the “Variable information” section.

Table 1. Distribution of observations, systems, and species by a selection of classification factors in *PROCEEDv6.2_RatesDB*.

Factor level	Observations	Systems	Species
Taxa			
Amphibian	23	5	5
Annelid	1	1	1
Arthropod	1095	20	17
Bird	1475	201	172
Fish	3671	1311	44
Mammal	686	98	63
Mollusc	45	4	4
Plant	2171	156	118
Reptile	96	5	4
design			
Allochronic	3435	1598	283
Synchronic	5828	203	145
genphen			
Genetic	3063	104	75
Phenotypic	6200	1697	353
trait_type			
behaviour	19	3	3
growth	332	17	12
otherLH	1518	59	34
othermorphology	3059	117	68
phenology	164	100	82
physio	634	37	24
response	78	14	12
size	3459	1454	193
disturbance			
Climate change	451	231	182
Hunt_harv	1365	794	31
Introduction	4979	129	88
Landscape change	375	38	25
Other	1529	551	55
Pollution	216	35	25
Response to introductions	348	23	22

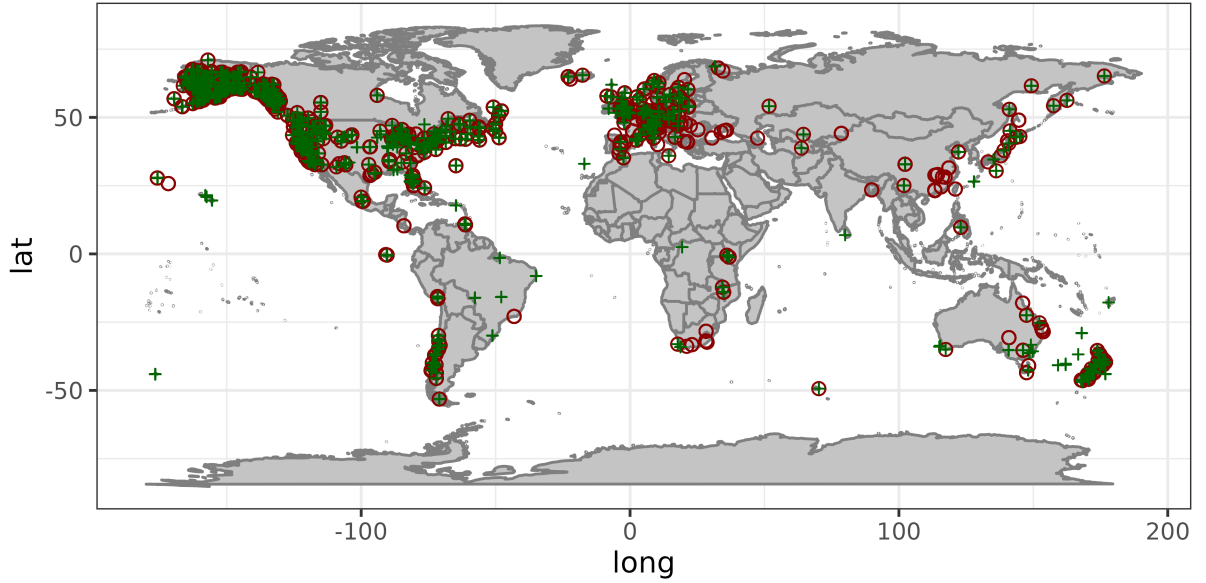


Figure 2. Geographic distribution of the populations recorded in PROCEED. Red circles show the position of the sample 1 (original population). Green crosses show the position of the sample 2 (derivative population).

Figure 3 shows the joint distribution of Darwins and Haldanes in PROCEED. According to previous definitions and considerations, Darwins cannot be computed if the trait is not measured in a ratio scale, or if the average mean of sample 1 (in a raw scale) is equal to zero, or if it is a missing value, or if the ratio between mean 2 and mean 1 (in a raw scale) is a negative value (414 records were excluded). Additionally, we did not compute Darwins if the trait measurement scale is unknown whether it is in the raw scale or any transformation (e.g. logarithmic) was applied. Only data in the raw or ln-transformed scale were included (89 additional records were excluded). Here, Darwins numerators were computed in the raw scale of the trait. If the scale of the trait was ln-transformed (ln: natural logarithm), the Darwins numerator was computed according to the Lynch (1990) approximation. That is, the natural logarithm of the average of the trait at a given time is approximated as the average of the natural logarithm of the trait plus the half of the variance of the natural logarithm of the trait:

$$\ln(\bar{x}_t) = \ln(\bar{x}_t) + 0.5 \cdot \sigma_{\ln(x_t)}^2$$

As a consequence, the records expressed in ln-transformed scale also requires the standard deviation to be computed. Thus, 202 additional records were excluded. Additionally, four records were assumed to be outliers and excluded because the ratio between the mean in sample 1 and sample 2 was larger than two orders of magnitude.

According to previous definitions and considerations, Haldanes cannot be computed if the standard deviation of sample 1 or 2, or any of the sample sizes are lacking (1620 records were excluded). Also, it cannot be computed if the generation time is unknown (274 additional records were excluded). Additionally, we assumed as outlier and excluded from the calculation of Haldanes any records where the standard deviation of any of the samples were zero or the coefficient of variation was smaller than 0.1 or larger than 1000 (79 additional records were excluded).

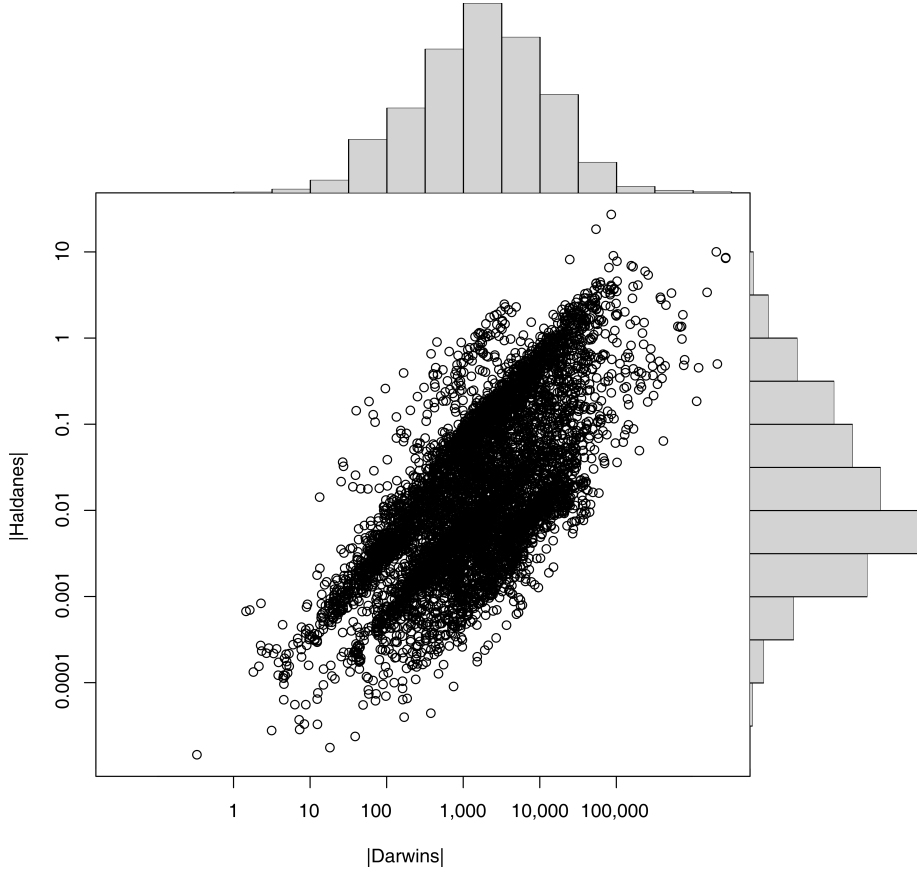


Figure 3. Distribution of absolute values for Darwins and Haldanes in PROCEED.

The dataset PROCEEDv6.2_TimeSeriesDB unpacks a subset of allochronic observations from the main dataframe. The time-series dataset compiles 655 time-series from 156 systems and 77 species. Table 2 summarizes the distribution of the series in a selection of the classification factors. Figure 4 shows the frequency distribution of the time-series length measured as both years and time-points. When the driver of change in a population was modified, the new situation was classified as a new system. The new system is linked to the previous situation via “released_sys”. For example, the system s445 (*Ovis canadensis* in Ram Mountain, Alberta, Canada when a hunting moratorium was placed starting in 1996) is the same population of system s122 (*Ovis canadensis* in Ram Mountain, Alberta, Canada when

hunting was allowed up to 1995). So, for the system s445 “released_sys”=“s122”. On the contrary, for the system s122 “released_sys”=NA (because s122 was not derived from any other system in the database). Because of this criterion, nine cases produced time series with only 2 time-points (belonging to 6 systems and 2 fish species). This is an exception to the general rule of including series with three or more time-points.

Table 2. Distribution of time-series, systems, and species by a selection of classification factors in PROCEEDv6.2_TimeSeriesDB.

Factor level	Series	Systems	Species
Taxa			
Amphibian	8	2	2
Bird	77	25	15
Fish	404	58	20
Mammal	140	58	27
Plant	22	12	12
Reptile	4	1	1
genphen			
Genetic	2	1	1
Phenotypic	653	155	76
trait_type			
growth	59	9	2
otherLH	78	10	8
othermorphology	176	63	32
phenology	22	15	6
physio	10	6	6
size	310	53	23
disturbance			
Climate change	132	68	30
Hunt_harv	322	48	19
Introduction	9	4	3
Landscape change	8	5	5
Other	171	24	13
Pollution	8	6	6
Response to introductions	5	1	1

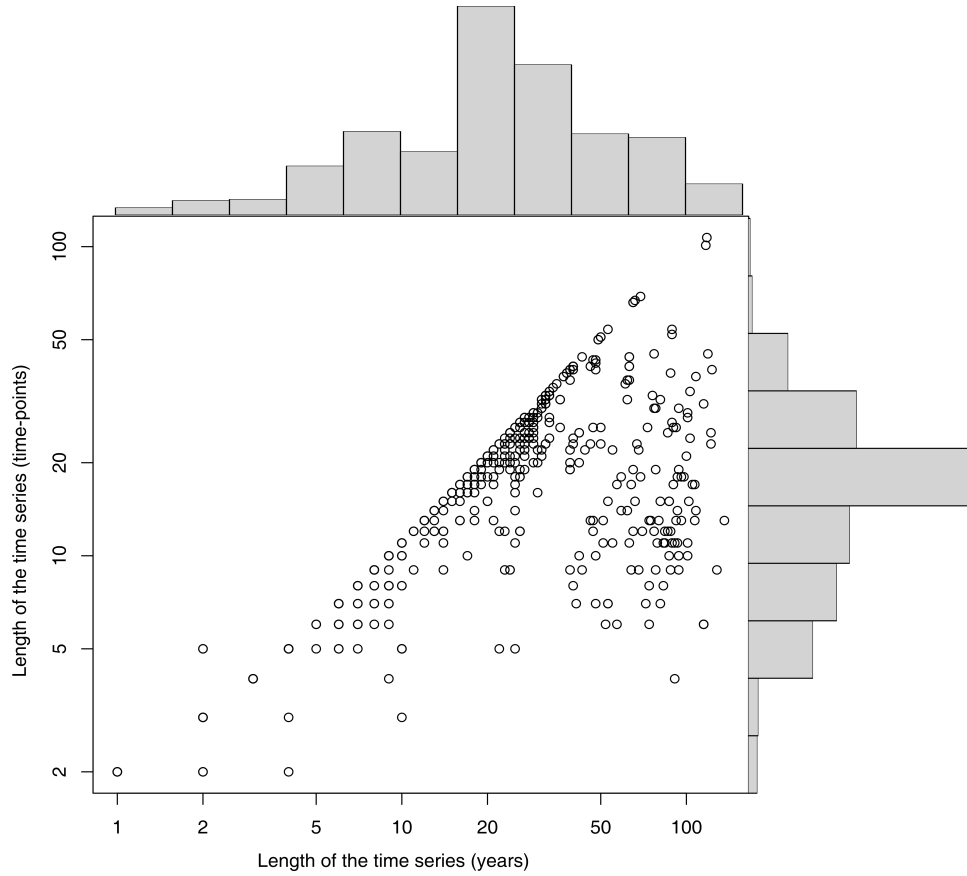


Figure 4. Length of the time series in PROCEED, showed both, in years, and in time points.

The datasets PROCEEDv6.2_species_genome and PROCEEDv6.2_species_GT are complementary tables with additional information about the genome size and generation time respectively of the species in the Rates and TimeSeries datasets (Figure 5). Rates, TimeSeries, and taxonomy tables are structured as a relationship database linked by primary and foreign keys (Figure 5). The complementary tables summarize information collected for each species and inform specific attributes of Rates and TimeSeries according to the criteria explained in the metadata.

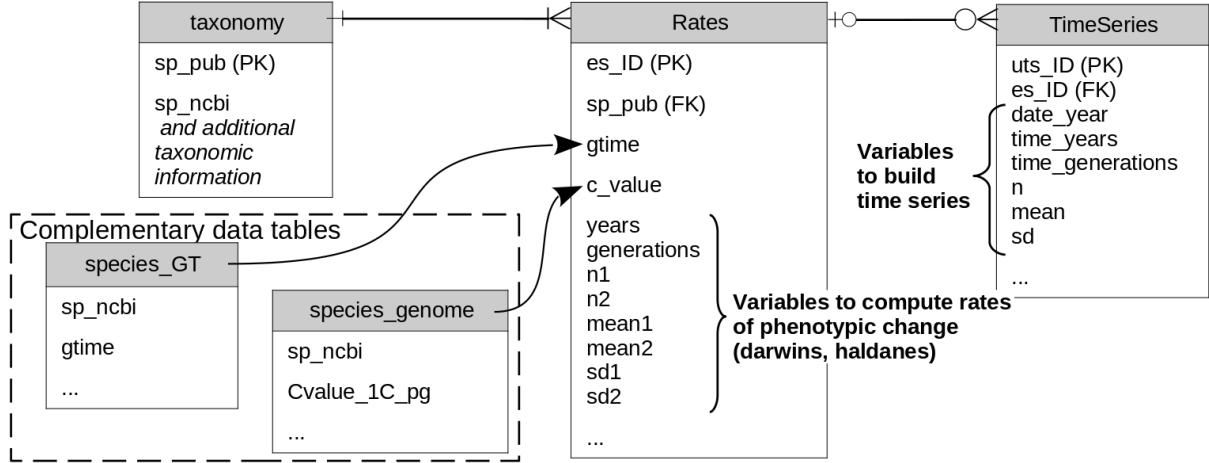


Figure 5. Entity Relationship Diagram of PROCEED. “Rates” inherits `sp_ncbi` and additional taxonomic information from the “taxonomy” entity. For each element in “taxonomy” there are from one to many elements in “Rates”. For each element in “Rates” there is one element in “taxonomy”. “TimeSeries” inherits attributes from “Rates”. For each element in “Rates” there might be from zero to many elements in “TimeSeries”. For each element in “TimeSeries”, there might be from zero to one element in “Rates”. Among the “Rates” attributes, are those required to compute Darwins and Haldanes. Among the attributes in “TimeSeries”, are those required to build the time series of phenotypic change in a study system. “...” denotes there are more attributes not listed here. See the metadata of each dataset for further details. PK: primary key; FK: foreign key. The complementary tables inform specific attributes of the “Rates” entity. “species_GT” and “species_genome” summarize all the generation time and genome size information collected for each species (`sp_ncbi`) respectively but there is not a PK for these two tables because more than one generation time or genome size can be recorded for each species.

PROCEED has been compiled to answer several questions about how phenotypic evolution is displayed in contemporary time scales and how different factors affect this process (its pattern, rate, and amount of change). Some of these factors have been introduced previously (factors listed in Table 1, for example) and have been analyzed with previous versions of PROCEED. Since its early versions, PROCEED has been the empirical support for a number of eco-evolutionary studies (Hendry & Kinnison 1999; Kinnison & Hendry 2001; Hendry et al. 2008; Darimont et al. 2009; Crispo et al. 2010; Westley 2011; Palkovacs et al. 2012; Gotanda et al. 2015; Alberti et al. 2017; Gorné & Díaz 2019; Sanderson et al. 2021, 2023). Some of these papers are fundamental contributions to the field of ecology and evolution. For example PROCEED showed that phenotypic change in the contemporary timescales is consistent with an abrupt initial change followed by a period of stasis or some degree of random changes (Kinnison & Hendry 2001; Gorné & Díaz 2019). The database also showed that the most rapid phenotypic changes tend to be in populations facing human disturbances such as pollution, urbanization, landscape change, invasive species, or hunting/harvesting (Alberti et al. 2017; Westley 2011; Sanderson et al. 2021; Darimont et al. 2009). In particular, pollution

and hunting/harvesting by humans caused more rapid changes compared to populations experiencing other types of disturbances (Darimont et al. 2009; Sanderson et al. 2021), and populations in urbanized systems have faster rates of change than populations in non-urbanized systems (Alberti et al. 2017). PROCEED also showed that the rates of change in life history traits are as fast or even faster than the rates of change in morphological traits (Kinnison & Hendry 2001; Gorné & Díaz 2019). The new version of PROCEED would allow us to dig deeper into human influences on rates of trait change as well as to address different questions like the relationship between some biological properties of the species (e.g. generation time, genome size, body size) and the patterns of phenotypic change. Future work can also look at different combinations of things known to affect trait change to see if they have synergistic or antagonistic effects on phenotypic change.

Here, as an example, we performed a similar analysis to the one by Sanderson et al (2021) to inquire about the effect of different human influences on the amount of phenotypic change they drive.

As previously reported by Sanderson et al (2021), we found that systems associated with pollution have the highest amounts of change and that systems associated with climate change have the lowest change (Figure 6). High rates of change due to pollution might be explained by the underlying genetic architecture of tolerance to pollutants (Macnair 1991; Oomen et al. 2020; Kardos & Luikart 2021).

The R code to reproduce Figure 6, as well as code for some basic handling of PROCEED and rates computing, is available in the file “PROCEEDv6.2_BasicCode.txt”.

PROCEEDv6.2_BasicCode has R code for some basic handling of PROCEEDv6.2 and examples to perform some basic analyses and graphics. See the metadata of each dataset for further details. Additionally to the stored PROCEED v6.2 in the stable repository (<https://doi.org/10.5683/SP3/NXSL3Q>), previous versions are available in the webpage of the database (<https://proceeddatabase.weebly.com/>) as well as in GitHub (<https://github.com/photopidge/PROCEED>).

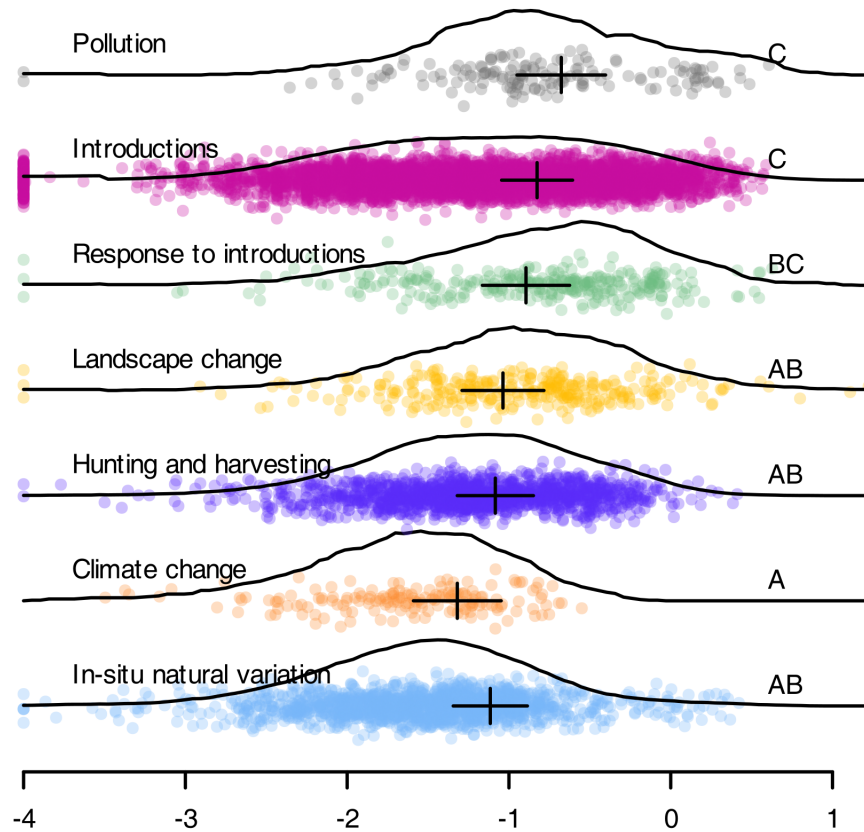


Figure 6. Amount of phenotypic change—in log10-transformed absolute darwin numerators—for six types of disturbances (pollution, response to introductions, hunting and harvesting, introductions, landscape change, climate change) and in situ natural variation (disturbance="Other"). Points show individual data, lines show smoothed data distributions, and crosses show linear mixed model estimates and 95% confidence interval for each disturbance type taking into account all the other variables in the database that might affect the observed amount of change. Letters indicate groups that differ according to Tukey's tests based on the fitted model.

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